surface mine plans and where required by the inspector, identified by a system of stakes or signs on the surface.

Elevating Work Platforms and Aerial Devices

Fall Arresting Devices

4.5.01 (1) Subject to subsection (5), where a person is exposed to the hazard of falling more than 3 m, a fall arresting device shall be used.

(2) The fall arresting device required by subsection (1) shall comply with the relevant design and performance requirements of
- (a) CSA Z259.1-95 “Safety Belts and Lanyards”, or
- (b) CSA Z259.2.1-98 “Fall Arresters, Vertical Lifelines and Rails”, or
- (c) CSA Z259.2.2-98 “Self-Retracting Devices for Personal RailArrest”, or
- (d) CSA Z259.2.3-98 “Descent Control Devices.”

(3) Safety belts, harnesses, lanyards and lifelines shall not be knotted or allowed to become knotted.

(4) When in use with a fall arresting device, a lifeline shall be anchored so that a person cannot fall, free of arrest, for more than 1220 mm, and it shall be connected to an object that is free from sharp edges and capable of resisting the force of an arrest.

(5) Subsection (1) does not apply to a person employed in shaft sinking where measures are in effect to provide equal or greater protection against falling.

Standards

4.5.1 Except as otherwise specified, all elevating work platforms, other than fire-fighting aerial devices, shall be designed, fabricated, operated, inspected, tested and maintained in accordance with the following standards:

(1) CSA Standard CAN/CSA-C225-00 “Vehicle-Mounted Aerial Devices,”

(2) CSA Standard CAN3-B354.1-M82 “Elevating Rolling Work Platforms,”

(3) CSA Standard CAN3-B354.2-01 “Self-Propelled Elevating Work Platforms,”

(4) CSA Standard CAN3-B354.4 “Boom-Type Elevating Work Platforms,”

(5) CSA Standard B335.1-1994 “Low Lift and High Lift Trucks,”

(6) manufacturer’s instructions, and

(7) other standards acceptable to the chief inspector.

Forklift Truck Platforms

4.5.2 Work platforms mounted on forklift trucks and not conforming to section 4.5.1 may be used for emergency or infrequent operations if they are secured to the fork carriage and meet the requirements of section 4.5.3.

Guard Rails

4.5.3 Elevating work platforms and all similar aerial platforms shall have

(1) standard guardrails on all open sides or be enclosed to provide equivalent protection (toe-boards may be omitted at the access openings),

(2) guards to protect the occupants from the elevating machinery, and

(3) signs clearly indicating the safe maximum working load.

Nondestructive Testing

4.5.4 All vehicle mounted aerial devices and self-propelled boom-type elevating work platforms shall have critical components non-destructively tested by a qualified person prior to their introduction to a minesite, and at intervals not to exceed 12 months.

Electrically Insulated Aerial Device

4.5.5 Any aerial device or component thereof that is stated by the manufacturer as being electrically insulated shall be tested annually to ensure that it meets the requirements of CAN/CSA-C225-00 “Vehicle- Mounted Aerial Devices”.

4.5.6 Any defects found during the tests required by sections 4.5.4 and 4.5.5, shall be repaired in accordance with instructions issued by a professional engineer before being returned to service.

Logbook

4.5.7 A logbook shall be maintained for each vehicle- mounted aerial device and self-propelled boom-type elevating platform in which any defect, operating difficulty, inspection, test, maintenance, modification and repair work is to be recorded. All entries shall be dated and signed by the person responsible for the work.

Mobile Equipment – Underground Mines

Mobile Equipment Underground

4.6.1 (1) The manager shall ensure that written procedures for the operation and maintenance of diesel-powered equipment are established.

(2) The fuel for a diesel engine shall conform with CAN/CGSB-3.16-99, “Mining Diesel Fuel,” Special- LS.

(3) A minimum of 0.06 cubic metre per second of ventilating air for each kilowatt of power of the diesel-powered equipment operating shall be circulated by mechanical means through every workplace where diesel-powered equipment is operating.

(4) Mobile diesel-powered equipment must not be left unattended while the engine is running.

(5) When the operator parks mobile diesel-powered equipment, there must be a cool down period as established by the OHSC or the manufacturer's specification.

(6) Subject to subsection (7), when the operator leaves a piece of diesel-powered equipment the master switch must be turned off.

(7) If autonomous or semi-autonomous tracked or rubber-tired mobile equipment is parked in a designated parking area and the engine or power source is turned off, the master switch may be left on.

General Requirements

4.6.2 (1) The manager must ensure that all underground mining mobile equipment introduced into service in an underground mine on or after the date this section comes into force complies with CSA Standard M424.0-2022 Underground mining mobile equipment – General requirements.

(2) The manager must ensure that underground mining mobile equipment introduced into service in an underground mine before the date this section comes into force complies with

(a) CSA Standard M424.0-2022 Underground mining mobile equipment – General requirements, or

(b) in the case of the equipment operating in

(i) an underground coal mine, with the general vehicle and machine requirements set out in CSA Standard CAN/CSA-M424.1-88, "Flame-Proof Non-Rail Bound Diesel-Powered Machine for Use in Gassy Underground Coal Mines", or

(ii) an underground mine, other than a coal mine, with the general vehicle and machine requirements set out in CSA Standard CAN/CSA-M424.2-90, "Non-Rail Bound Diesel-Powered Machines for use in Non Gassy Underground Mines".

Trackless Diesel Powered Equipment

4.7.1 (1) The manager must ensure that trackless diesel-powered equipment, made for use in

(a) underground coal mines, complies with CSA Standard M424.1-2022 Flameproof non-rail-bound diesel-powered machines for use in gassy underground coal mines, and

(b) underground mines other than coal, complies with CSA Standard M424.2-2022 Diesel-powered machines for use in non-gassy underground mines.

(2) Subsection (1) (a) does not apply to trackless diesel-powered equipment, made for use in underground coal mines, if

(a) the equipment is not used for cutting, digging and loading coal, and

(b) the manager, prior to the use of the equipment,

(i) makes health and safety procedures relating to the equipment,

(ii) updates the Mine Health and Safety Program, as set out in section 1.6.9 (1) (c), in relation to the health and safety procedures, and

(iii) submits a copy of the health and safety procedures to the chief inspector.

Rubber-tired, Self-propelled Underground Mining Machines

4.7.2 The manager must ensure that rubber-tired, self-propelled underground vehicles used in underground mines comply with CSA Standard M424.3-2022 Braking performance – Rubber-tired, self-propelled underground mining machines.

Self-propelled, Electrically driven, Non-rail-bound Mobile Machines

4.7.3 The manager must ensure that self-propelled, electrically driven, non-rail-bound mobile machines used in underground mines, other than coal, comply with CSA Standard M424.4-2022 Self-propelled, electrically driven, non-rail-bound mobile machines for use in non-gassy underground mines.

Off-road Vehicles (ORVs)

4.8.1 [Renumbered as 4.3.10]

4.8.2 [Renumbered as 4.3.11]

4.8.3 [Renumbered as 4.3.12]

Prohibitions

4.8.4 The manager must ensure that all-terrain vehicles (ATVs) and all-terrain cycles (ATCs) are not used on the mine site.

Manager's Responsibilities

4.8.5 (1) The manager must ensure that the operator of an off-road vehicle (ORV) is

(a) under the supervision of a qualified person, and

(b) trained for the work being performed using the off-road vehicle (ORV).

(2) The manager must ensure that procedures for the safe operation of off-road vehicles (ORVs) used for mining activities are developed, implemented and followed.

Supervisor's Responsibilities

4.8.6 (1) The supervisor must ensure

(a) that, prior to operating an off-road vehicle (ORV), the operator has completed training on

(i) the specific type of off-road vehicle (ORV) that the operator will be using, and

(ii) the terrain on which the operator will be using the off-road vehicle (ORV), and

(b) that an operator of an off-road vehicle (ORV) is competent and authorized to do so.

(2) The supervisor must ensure that each off-road vehicle (ORV) is maintained in accordance with the operator's manual supplied by the manufacturer.

Worker's Responsibilities

4.8.7 (1) A worker must not operate an off-road vehicle (ORV) unless the worker

(a) has completed the training required under sections 4.8.5(1)(b) and 4.8.6(1)(a), and

(b) is authorized by a qualified person.

(2) When operating an off-road vehicle (ORV), the worker must

(a) maintain full control of the off-road vehicle (ORV), and

(b) operate the off-road vehicle (ORV)

(i) in a safe manner at all times,

(ii) only within its intended work purpose, and

(iii) in accordance with the training referred to in subsection (1) (a).

Transporting

4.8.8 (1) The manager must ensure that transporting an off-road vehicle (ORV) is done in a safe manner, including ensuring that the off-road vehicle (ORV) is

(a) secured to a trailer while in transit, and

(b) not transported in or on the back of a truck.

(2) The manager must ensure that loading or unloading an off-road vehicle (ORV) on or off a trailer is done in a safe manner, including ensuring that

(a) a loading ramp is secured to the trailer,

(b) the load rating of the loading ramp is not exceeded, and
(c) the loading ramp is wide enough, is at a suitable angle and has an appropriate grip surface for the off-road vehicle (ORV) tires or tracks to properly grip and not slip.

Use of Off-road Vehicle

4.8.9 (1) Despite section 6.19.2, the manager must ensure that
(a) a logbook system is provided for each off-road vehicle (ORV), and
(b) before an off-road vehicle (ORV) is used, it is inspected by the operator who, in the logbook,
(i) notes the condition,
(ii) notes any unsafe conditions, and
(iii) in the case of a qualified person making repairs, notes any repairs made and if the off-road vehicle (ORV) is safe to operate.

(2) The manager must ensure that before an off-road vehicle (ORV) is used, it is outfitted with
(a) a whip antenna that is fitted with a flag and lamp high enough to be visible to the drivers of all vehicles, or
(b) a flashing light mounted above the cab of the off-road vehicle (ORV).

(3) The manager must ensure that any modification to an off-road vehicle (ORV) that could impact its structural integrity or stability is certified by a professional engineer before it is used.

(4) The manager must ensure that the use of an off-road vehicle (ORV) with an operator cab is only permitted if the windows and doors of the operator cab are kept installed and in good condition.

(5) The manager must ensure that when an off-road vehicle (ORV) is being used, the operator and passengers, if any, wear
(a) appropriate personal protective equipment, as determined by a risk assessment,
(b) an off-road vehicle (ORV) helmet that meets the requirements set out in section 22 of the Off-Road Vehicle Regulation, B.C. Reg. 193/2015, other than while underground in an underground mine,
(c) a seatbelt, except in the case of a snowmobile, and
(d) any personal protective equipment recommended in the owner's manual that does not conflict with the requirements of paragraphs (a) to (c) of this subsection.

(6) The manager must ensure that the operator's manual of an off-road vehicle (ORV) is kept with the off-road vehicle (ORV) and maintained in a readable condition.

Mobile Equipment

Rubber-tired and Tracked Mobile Equipment Maintenance

4.9.01 (1) The manager shall ensure that brake and steering systems on tracked or rubber-tired mobile equipment are maintained in safe operating condition through a regularly scheduled preventative maintenance program.

(2) Maintenance on braking and steering systems on tracked or rubber-tired mobile equipment shall be performed by or under the supervision of a certified person.

(3) Records of the maintenance under subsection (1) must be readily available for review by an inspector upon request.

Back-up Alarm

4.9.1 (1) The manager shall ensure that every unit of rubber tired mobile equipment in regular use at a mine in excess of 7000 kg gross vehicle weight is equipped with a "back-up" alarm which meets the requirements of SAE J-1446 and SAE J-994. Underground mobile equipment designed for bi-directional use shall be excluded from this requirement.

(2) The alarm shall be clearly audible above the background noise at the workplace and it shall continue as long as the equipment is moving in reverse.

(3) Mobile equipment operating in a mine may use a visual back-up alarm in place of an audible back-up alarm providing that the visual back-up alarm complies with the following criteria:

(a) the light shall be clearly visible above the usual daily light levels,

(b) actuation of the alarm shall be completely independent of the operator, and

(c) the light shall flash as long as the vehicle transmission is in the reverse position.

Transmission Interlock

4.9.2 Every unit of mobile equipment having a fluid drive transmission shall be equipped with an interlocking system, maintained in working condition, to prevent the unit from being started and put into motion unless the transmission selector is returned through the neutral position.

Fire Extinguisher

4.9.3 Every vehicle shall carry at least one fire extinguisher of adequate size and of the proper type.

Vehicle Requirements

4.9.4 (1) All rubber tired mobile equipment over 7000 kg gross vehicle weight shall have a minimum of two wheel chocks which shall be used whenever necessary.

(2) All mobile equipment shall have, maintained in working condition, the following:

(a) a firmly secured seat for the operator and any authorized passenger, well maintained in a comfortable, shock absorbing condition;

(b) approved safety seat belts;

(c) suitable clearance lights and reflectors;

(d) if an operator cab was provided as part of the original equipment package, or subsequently fitted, windshields, side and rear windows, and rear vision mirrors maintained to provide clear visibility, and the glazing material shall meet the specifications of ANSI Standard Z26.1 - 1990 “Safety Glazing Materials for Glazing Motor Vehicles.”

(3) The manager must ensure that mobile equipment and non-self-propelled barges or dredges performing work where drowning may be a risk are equipped with a secondary egress that is designed for the purpose and does not impede the safe exit of the operator.

4.9.5 A vehicle, other than a vehicle used directly for production in an operating open pit, shall, additional to that equipment required for highway driving, be equipped with

(1) a whip antenna that is fitted with a flag and lamp high enough to be visible to the drivers of all production vehicles, or

(2) a flashing light mounted above the cab of the vehicle.

Modifications Respecting Mobile Equipment Where Drowning May Be a Risk

4.9.5.1 The manager must ensure that, if one or more parts of mobile equipment or non-self-propelled barges or dredges performing work where drowning may be a risk have been modified, and the modification may affect the safe operation of the mobile equipment or non-self-propelled barge or dredge, a qualified professional approves the modification before the mobile equipment or non-self-propelled barge or dredge is used by workers.

Fire Suppression For Underground Mines

4.9.6 (1) All mobile equipment capable of containing more than 25 L of flammable fluids and used in underground mines shall be fitted with a manually activated multi-nozzle fire suppression system. Manual activation of the system shall be possible from each side of the machine and at the operator’s station.

(2) Actuation of the fire suppression system shall also cause engine shutdown.

(3) Electric hydraulic diamond drills shall be equipped with
(a) a deflector shield between the hydraulic pump and the motor, and
(b) a temperature probe in the hydraulic oil, which will shut down the drill if oil temperature reaches 65 degrees Celsius.

Transportation of Persons

4.9.7 The manager shall ensure that vehicles regularly or primarily used as personnel carriers are suitable for the intended purpose.

Personnel Carriers and Hazardous Materials

4.9.8 (1) Every personnel carrier shall be inspected each day by an authorized person before being used to transport persons, and if any defects are found which could affect the safe operation of the vehicle they shall be corrected before it is used.

(2) Where it is necessary to transport hazardous materials on a personnel carrier at the same time as persons, the materials shall be in an isolated compartment, adequately ventilated and drained, and accessible only from outside of the vehicle. If the compartment is inside the vehicle, it shall be separated from the passenger section by an approved firewall.

Data Plate

4.9.9 (1) No truck or loader shall be operated within a mine unless there is affixed to the inside of the cab, at a place clearly visible to the operator, a nameplate of the vehicle manufacturer showing the serial number and maximum rated load capacity of the vehicle.

(2) Where a truck, scraper or any other rubber tired vehicle, has a gross vehicle weight in excess of 45,000 kg, the name plate referred to in subsection (1) shall also show the maximum allowable speeds at the rated gross vehicle weight that the retarder and/or braking system is designed to accommodate when traveling on
(a) the maximum downgrade on which the vehicle is designed to operate, and
(b) not less than 2 other specified slopes between such maximum downgrade slope and level ground.

Braking and Steering Modifications

4.9.10 Every truck, loader or other rubber tired vehicle having a manufacturer's gross vehicle weight in excess of 45,000 kg shall have any modification affecting the braking, steering or resulting in an increase to the gross vehicle weight be approved by the chief inspector.

Rollover Protective Structures

4.9.11 (1) A loader, grader, scraper, tractor, compactor roller, skidder, rough terrain fork lift, bulldozer, and off-highway haul truck introduced into service after July 1, 1991, shall be equipped with a rollover protective structure that conforms with any of the following standards

(a) Society of Automotive Engineers, Standard SAE-J1040 April 1988 "Performance Criteria for Rollover Protective Structures (ROPS) For Construction, Earthmoving, Forestry and Mining Machines,"

(b) "Earthmoving Machinery Rollover Protective Structures – Laboratory Tests and Performance Requirements ISO 3471, or

(c) Canadian Standards Association, Standard B352.095 "Rollover Protective Structures (ROPS) for Agricultural, Construction, Earthmoving, Forestry, Industrial, and Mining Machines."

(2) Where the mobile equipment referred to in subsection (1) is designed mainly for underground use,

(a) a ROPS designed by a professional engineer shall conform to Clause 6 of CSA B352-M1980 but does not require physical testing, or

Falling Object Protective Structures (FOPS)

(b) a FOPS (Falling Object Protective Structures) canopy may be substituted for a ROPS canopy provided it is specifically approved by the chief inspector.

(3) Side-by-side utility task vehicles (UTVs) must have a ROPS that conform to OSHA29 BFR1928.53 certified ROPS or Canadian Standards Association, Standard B352.095 or another standard as accepted by the chief inspector.

4.9.12 (1) The following information shall be permanently marked on rollover protective structures

(a) the name and address of the manufacturer or the registered professional engineer who certified the ROPS,

(b) the model,

(c) the make, model, and series of the machine for which the rollover protective structure is designed, and

(d) where the ROPS has been designed and constructed in accordance with section 4.9.11 (2), the notation "Underground Use Only" shall be added.

(2) Seat belts of a type conforming with the recommended Practice of the Society of Automotive Engineers SAE-J-386 "Operator Restraint Systems for Off-Road Work Machines" shall be installed in all mobile equipment fitted with a rollover protective structure and the operator of the equipment shall wear the seat belt when operating the equipment.

Rollover Protective Structures ROPS

4.9.13 (1) No addition, modification, welding or cutting of a ROPS shall be permitted except in accordance with the instructions of a registered professional engineer and all modified ROPS shall be permanently marked to show the modifications made, the date of recertification and the name and address of the certifying engineer.

(2) A registered professional engineer may re-certify a damaged, deformed or deteriorated ROPS, after repairs have been made, if the registered professional engineer is satisfied that the structural integrity has not been impaired.

(3) ROPS shall be designed and fitted so as not to restrict the operator's field of vision

Tires and Rims

4.9.14 (1) The manager shall ensure that procedures are in place for the inspection and any work on tires and rims of equipment.

(2) A tire shall not be installed on any damaged, broken, bent or heavily rusted rim assembly and mismatched parts of rims and wheels shall not be used.

(3) No person shall work on tires and rims unless qualified.

Securing of equipment

4.9.15 No person shall place themselves under the raised box of a dump truck, bulldozer blade, scraper blade, loader bucket or similar equipment unless it is securely and adequately blocked or otherwise secured independently of normal operating controls with suitable equipment that is designed for that purpose.

Brakes

4.9.16 (1) Mobile equipment must have braking systems meeting the requirements of the following applicable standard, or other standard acceptable to the chief inspector:

(a) Society of Automotive Engineers (SAE) Standard J/ISO3450 JAN 98, Earthmoving Machinery – Braking Systems of Rubber-Tired Machines – Systems and Performance Requirements and Test Procedures;

(b) Society of Automotive Engineers (SAE) Standard J1026 APR 90, Braking Performance – Crawler Tractors and Crawler Loaders;

(c) Society of Automotive Engineers (SAE) Standard J1472 JUN 87, Braking Performance – Roller Compactors;

(d) ANSI Standard ANSI/ASME B56.6-1992, Safety Standard for Rough Terrain Forklift Trucks;

(e) ANSI Standard ASME B56.1-1993, Safety Standard for Low Lift and High Lift Trucks.

(2) When required by the inspector, an independent means of retardation shall be provided in addition to the requirements of subsection (1).

Parking Brake

4.9.17 Every motor vehicle shall be equipped with an effective, mechanically activated parking brake, the holding power of which is not affected by loss of fluid or air pressure from the braking system.

Common Components

4.9.18 On any vehicle where components that apply the service brakes and the emergency brakes are common, the components shall be arranged so that a failure in a common component does not reduce the capability of one of the systems to stop the vehicle safely.

Brake Testing (Annual)

4.9.19 (1) Trucks having a manufacturer's rated gross vehicle weight in excess of 45,000 kg shall be subjected to downgrade braking tests such that at least once a year 50% of the fleet is tested, and every truck in the fleet is tested at intervals not exceeding 3 years. The tests shall be as follows:

(a) the vehicle shall be loaded to the manufacturer's maximum permissible gross vehicle weight,

(b) the tests shall be conducted on a well-graded hard packed downgrade slope of uniform grade of between 8% and 10% or where there is no downgrade slope satisfying this condition, tests shall be conducted on the maximum downgrade slope over which the trucks are operated,

(c) auxiliary retarding devices shall not be used during the brake tests, and

(d) stopping distances shall be measured from the initial point of application of the service brakes to the final stopping position, from an initial speed of 40 km/h, and the minimum acceptable brake performance shall be as follows:

Initial Speed 40 km/h

Grade	Stopping Distance
10%	84 m
9%	76 m
8%	68 m (2)

Where a vehicle fails to meet the minimum brake performance standards specified under subsection (1) (d), it shall be removed from service until corrective measures have been taken and the vehicle meets the standard.

Auxiliary Steering

4.9.20 (1) If wheeled mobile equipment having rated speed capability exceeding 20 km/h depends on engine power for steering and power failure will prevent the vehicle from being steered manually, a supplementary system shall be provided to enable the operator to steer to a controlled stop.

(2) The supplementary steering system required by subsection (1) shall meet the requirements of Society of Automotive Engineers (SAE) Standard J151 1 ISO5010 FEB 94, Steering For Off-Road, Rubber-Tired Machines.

(3) When hydraulic accumulators are used to provide power to supplemental steering systems, they shall depressurize when the engine is deliberately shut down.

Automatic Engine Shutdown

4.9.21 Where automatic engine shutdown devices are employed on vehicles, audible and visual alarms shall be installed in the operator's cab to pre-warn the operator that an automatic engine shutdown is imminent.

Railways

Trains – Fitments

4.10.1 (1) All trains shall be equipped with suitable tail lights and clearance lights.

(2) If trains are required to reverse, they shall be equipped with a suitable beam or flashing tail light and if they are to be reversed frequently and for lengthy distances, a procedure shall be followed which is acceptable to the inspector.

(3) Every trolley locomotive shall be operated with the trolley pole in the trailing position, unless there is no room to reverse the pole, in which case speed shall not be more than walking speed.

(4) A car shall not be pushed by a locomotive if material extends beyond the length of the car.

(5) Each locomotive shall be equipped with suitable fire extinguishers.

(6) Nothing shall be placed on the top of a locomotive unless it is necessary for its operation and, in that case, it shall be adequately secured, must not extend beyond the ends or sides of the locomotive or into the cab area, and must not obstruct the operator's view.

4.10.2 An engine, locomotive, or trolley car used for hauling material shall be equipped with

(1) properly maintained headlights and clearance lights, and

(2) an audible warning system, which shall be sounded by the operator to warn persons when they may be endangered by movement of the locomotive or train, and whenever the locomotive or train is about to move.

Control Levers

4.10.3 Every storage battery and trolley haulage locomotive shall be equipped with a "deadman" control switch and with a control lever so installed that the lever cannot be removed when the power is on.

Unattended Locomotives

4.10.4 (1) No person shall leave an electric haulage locomotive unattended unless the brakes have been set, the control lever placed in the park position, and the main switch placed in a non-operating position.

(2) Subsection (1) does not apply if the locomotive is on automatic control and approval has been obtained from the chief inspector.

(3) When operated by remote control or by an automated system, it shall be arranged that in the event of any failure of the control or system, the locomotive and cars will be brought to a stop immediately.

Lock-Out Procedures

Lock-Out Procedures

4.11.1 (1) The manager shall develop a lock-out procedure which includes but, is not limited to, the requirements of section 4.9.15 and sections 4.11.2 to 4.11.7 inclusive, and the manager shall ensure that all persons required to lock-out machinery or equipment are adequately trained in the procedure and that a written copy of it is made available to them.

(2) Notwithstanding subsection (1) where a large number of persons is involved, the manager may develop a lock-out procedure, acceptable to an inspector that modifies the requirements of sections 4.11.4 and 4.11.5.

(3) The lockout procedure developed in section 4.11.1(1) should address the event that a lock has not been removed and the shift has ended.

Power to Be Cut Off

4.11.2 (1) Before any work is performed on electrical machinery or equipment, the main power source shall be disconnected, locked-out and tagged.

(2) Where machinery or equipment to be worked on is powered by a source other than electricity, the power supply shall be shut off, locked-out and tagged. A means shall be provided to safely release the stored energy from the machinery or equipment before any work begins. If a valve in a pipe could leak and allow water, steam, compressed air or other potentially hazardous substance to reach persons working on the machinery or equipment, the pipe shall be blanked off, or otherwise isolated, in accordance with the lockout procedure.

Locks and Tags

4.11.3 (1) Locks and tags shall be issued to each person who works on machinery or equipment that has to be locked out.

(2) A lock issued to any person shall only be capable of being opened by that person's key except that where a number of locks are issued to a person for the person's sole use, they may be mastered to a single key.

(3) Tags issued to individual persons shall contain space for the recording of the person's name, the type of work being performed, the date and time the work was started and the name of the supervisor in charge.

Affixing and Removal of Locks and Tags

4.11.4 (1) Each person who works on machinery or equipment requiring to be locked out shall be responsible for affixing the person's own lock and tag to the lockout device and for removing them on the completion of the person's work.

(2) The person who affixes the first lock in the lockout procedure shall, before the work begins, attempt to start the equipment or machinery to ensure that it is properly locked out.

(3) A lock shall only be removed by the person who affixed it to the lockout device.

Overlap Between Shifts

4.11.5 (1) When machinery or equipment is locked out, employees coming on shift shall place their own locks on the lockout device before the employees going off shift remove theirs.

(2) A supervisor may lockout the machinery or equipment during a shift change to allow employees going off shift to remove their locks.

Procedure Before Work Recommences

4.11.6 When work is completed on locked out machinery or equipment, and before any locks or tags are removed, all guards and other safety devices shall be replaced.

Watchperson's Responsibilities

4.11.7 In the event that the machinery or equipment cannot be locked out, a tag shall be affixed and a watchperson shall be posted at a location where the watchperson can prevent anyone from re-energizing the power supply and starting the machinery or equipment. The watchperson shall have no other duties at the time and the watchperson shall remain at the watchperson's post until told by the supervisor that the watchperson may leave.

Lifting Devices – General (Bridge and Overhead Traveling Cranes, Monorails and Underhung Cranes)

4.12.1 Electrical bridge and trolley conductors shall be located or guarded to prevent accidental contact by persons.

4.12.2 Each hoist shall have a device to prevent hook travel beyond the safe upper limit at all designed speeds.

4.12.3 Where electrically powered hoisting equipment is operated from cabs, means shall be provided in the cab for the operator to safely disconnect the main power supply under any load condition.

Manually-Operated Hoists and Winches

4.12.4 (1) A hand-operated hoist shall have a ratchet and pawl mechanism, load brake, or other means of safely holding the load at any height.

(2) Crank operated winches, not fitted with automatic load brakes, shall have a means of preventing the crank handle from slipping off the crank shaft while hoisting. On such winches, under free wheel conditions, the crank handle shall be removed before the load is lowered.

Mobile Cranes

4.12.5 Mobile cranes shall be operated with their turntables level, except as permitted by the manufacturer. Level indicating devices shall be provided and used for this purpose.

4.12.6 (1) Outriggers, when deployed to meet load capacity chart requirements, shall be fully extended and secured against retraction. Outrigger beams shall be marked to indicate their fully extended position and jacks shall be extended sufficiently to bear the whole weight of the crane.

(2) Outrigger floats shall be secured to the outrigger jacks when in use.

4.12.7 Mobile cranes shall be securely chocked on firm ground before any hoisting or lowering begins.

4.12.8 Where required by an inspector, any mobile crane that has a lifting capacity exceeding 10 t shall be equipped with a load weight indicator.

Powered Hoists and Winches

4.12.9 Air operated hoists and winches shall be supplied with air at sufficient pressure to ensure safe operation. Inadvertent disconnection of air supply hoses shall be prevented.

Rigging and Slings – General

4.12.10 All rigging and slinging work shall be conducted by, or under the supervision of, qualified persons who are familiar with all aspects of the work and with the proper signals.

Cranes General

4.12.11 A suitable non-destructive test shall be made by a person certified in accordance with the applicable standards of the Canadian General Standards Board 48-GP-4M, 48-GP-7M, 48-GP-8M, and 48-GP-13M on all rigid load carrying components of mobile cranes and bridge cranes greater than 10,000 kg capacity and any other cranes when required by the inspector, before being put in to service and at subsequent intervals not exceeding 12 months.

Automotive Lifts and Other Vehicle Supports

4.12.12 (1) An automotive lift or hoist shall meet the requirements of ANSI Standard ANSI/ALI B 153.1-1990, American National Standard for Automotive Lifts – Safety Requirements for the Construction, care, and Use, or other equivalent standard.

(2) A shop crane, jack, axle stand, ramp or other type of vehicle support shall meet the requirements of the applicable section of ANSI Standard ASME PALD-1993, Portable Automotive Lifting Devices, or other equivalent standard.

(3) The rated load capacity shall be marked on each automotive lift, hoist, axle stand, ramp, or other vehicle support and must not be exceeded.

(4) If a device listed in subsection (3) is modified, or if the manufacturer's rated load capacity is not known, the rated load capacity shall be established by a professional engineer.

Raise Climbers

4.13.1 (1) A raise climber shall not be put into service unless all critical, load-bearing components of the complete assembly and accessories have been inspected and tested non-destructively by approved methods.

(2) Subsequent to installation, the tests required by 4.13.1 (1) shall be done at intervals not exceeding 12 months.

(3) A raise climber shall be:

(a) designed, maintained and operated in accordance with good engineering practice; and

(b) built and installed according with the design.

Identification Plate

4.13.2 Every raise climber shall have a durable and legible identification plate showing

(1) the name of the manufacturer, the date of manufacture, the model number, and a serial number, and

(2) the maximum allowable speed and the maximum allowable load ratings as certified by the manufacturer or a professional engineer.

Brakes Required

4.13.3 (1) Raise climbers shall be equipped with at least 2 separate and independently operated service brakes, each capable of safely stopping and holding the conveyance under all rated conditions of load and speed.

(2) An automatic overspeed brake shall be installed that is capable of bringing the conveyance to a safe stop under any rated load condition from a predetermined overspeed.

(3) Each brake or braking system shall be capable of being tested independently.

(4) Where electro-mechanical brakes are installed, they shall be actuated immediately if the power supply to the climber is interrupted.

4.13.4 Repealed.

Controls

4.13.5 An emergency switch shall be provided in the cab of every electrically operated raise climber that will cut off the power supply to the drive motors if the main control contactor fails to open, or in any other emergency situation.

Electric Fittings

4.13.6 (1) All electrical equipment, including switches, connectors, wiring and cables shall be designed, installed and weather-proofed to ensure the safety of the raise climber under any operating conditions.

(2) An electrically powered raise climber shall not be operated in excess of 750 v, and shall be protected by a ground fault system.

Raise Climber Platforms

4.13.7 (1) No one is to ride on the platform of a raise climber while traveling to the face, and where the inclination of a raise is greater than 60 degrees, a substantial cover shall be provided and shall be used when so directed by the supervisor or at the discretion of the worker.

Anchoring of Work Platform

(2) A work platform shall be securely anchored to the raise climber guides or rails while persons are working on the platform.

(3) Unless otherwise approved by the chief inspector, only one raise climber conveyance shall be installed in any raise.

Bolts to Be Torqued

4.13.8 All bolts used in the assembly of a raise climber shall be of the proper grade and size, as specified by the manufacturer. The bolts shall be correctly torqued in accordance with the manufacturer's specifications and no defective or damaged bolts shall be used. All exposed bolts, and other components, shall be adequately protected against falling rock.

4.13.9 Only racks and pinions supplied by the raise climber manufacturer, or certified by a professional engineer, shall be used.

Modifications to Be Approved

4.13.10 Modifications designed to increase the capacity or speed of a raise climber, or modifications that could affect the load carrying capacity of a raise climber or a work platform, shall not be made unless approved by the manufacturer or a professional engineer.

Communication Required

4.13.11 An effective means of communication shall be provided between a raise climber conveyance and the base from which it operates.

Limit of Travel Stops

4.13.12 (1) Whenever a raise climber is operating, the end of the track on which it travels shall be provided with a stop block to prevent the conveyance from being taken beyond the track.

(2) All permanent, electrically driven raise climber installations shall be provided with devices which will automatically stop the conveyance at the upper and lower limits of travel.

Qualified Persons to Maintain Equipment

4.13.13 (1) The manager shall appoint qualified persons to establish mechanical and electrical maintenance schedules for each raise climbing installation, and to ensure that these are carried out in accordance with the manufacturer's recommendations and the provisions of the code.

Records to Be Kept

(2) A record of all prescribed tests, inspections, repairs, and maintenance work carried out, and any defects, damage or problems noted during the inspections or tests, shall be entered in a Raise Climber Logbook which shall be kept at the mine.

Logbook

(3) All entries in the logbook shall be dated and signed by the person who carried out the work, and the supervisor having charge of the installation shall check and countersign the entries at least once each week.

Daily and Weekly Checks

4.13.14 (1) A mechanic shall check an operating raise climber each day to ensure it is operating normally.

(2) The brakes and controls of a raise climber shall be tested prior to first being used during a workshift. The calibration indicator on the overspeed brake shall be checked daily.

(3) Before recommencing operations after a prolonged shutdown, a full inspection of the complete installation shall be carried out by authorized persons.

Fire Extinguisher

4.13.15 Every raise climber shall be provided with a suitable fire extinguisher.

Emergency Procedures

4.13.16 (1) Means shall be readily available to enable persons to descend, if trapped in a raise climber conveyance up a raise, in the event of a power failure or other emergency situation.

(2) The manager shall ensure that operators of raise climbers and, any other persons who would be involved in an emergency descent of trapped persons, are trained in the use of the emergency equipment.

Temporary Work Platforms – General

Scaffold Requirements

4.14.1 Where scaffolding or temporary work platforms are used, they shall be constructed and maintained in accordance with WCB Occupational Health and Safety Regulation 13.

Persons Not Allowed to Ride

4.14.2 No person shall ride on loads, slings, hooks, work platforms, or other similar equipment, unless specifically authorized by the manager, and the authorization shall not be granted for reasons of expediency or personal convenience, but only where it is necessary to the work process, and only when the provision of the normal means of access or conveyance is impracticable.

Requirements

4.14.3 Unless provided elsewhere in the code, work platforms suspended from hoisting equipment shall

(1) be designed by a professional engineer and copies of the design and fabrication drawings shall be kept at the mine,

(2) be equipped with standard guardrails and toe boards on all open sides or be enclosed to give equivalent protection,

(3) be clearly marked with an identification number, the weight of the platform and rigging and the safe working load of the platform,

(4) have supporting hooks and shackles latched or moused to prevent dislodgement, and

(5) not use spreader bars between the load hook and the platform.

Requirements

4.14.4 Work platforms suspended from hoisting equipment shall

(1) be such that the weight of the platform and rigging plus the safe working load of the platform shall not exceed 1/4 of the safe working load of the hoisting equipment,

(2) have all suspension slings and attachments rigged to provide a safety factor of not less than 10. The platform shall be designed with a safety factor of 4,

(3) be suspended from cranes having power booms or fixed booms and from hoisting gear capable of lowering under power, and hoisting and lowering speeds shall be kept as low as practicable and lowering of persons shall only be performed under power,

(4) have power hoists and winches of a type approved for such purpose by the manufacturer,

(5) have dogs in the hoisting equipment drive mechanism secured against inadvertent disengagement,

(6) have a designated person on the platform direct all movement of the platform, and

(7) have persons on the platform wear safety harnesses with lanyards secured to a substantial anchorage on the platform.

Crane or Hoist Operations

4.14.5 A crane or hoist shall be operated by a qualified and authorized person whenever a person is on a suspended platform and the operator shall remain at the controls while the platform is suspended.

Boom Cranes

4.14.6 (1) Telescoping-boom cranes used to suspend a work platform shall incorporate an "Anti-Two Block" device referenced in CSA Standard Z150-1974 "*Safety Code for Mobile Cranes*."

(2) Non-telescoping boom cranes used with suspended work platforms shall be equipped with an anti-two block warning device which alerts the crane operator before a two block situation occurs. The warning shall be activated when the load hook from which the platform is suspended is within 2 m of the boom sheave.

Inspection

4.14.7 Every suspended work platform shall be regularly inspected by an authorized person, as often as necessary to ensure that it can safely handle its rated load, and if any defect or doubtful condition is noted, the platform shall not be used until

(1) repairs or corrective action has been taken, and

(2) the load carrying components have been nondestructively tested and the platform re-inspected.

Logbooks

4.14.8 A logbook shall be maintained for suspended work platforms in which all tests, inspections, maintenance work and repairs shall be recorded. All entries shall be dated and signed by the person responsible for the work.

Construction – General

4.15 Construction shall be carried out in accordance with instructions of a qualified person and in compliance with WCB Occupational Health and Safety Regulations, Part 20.

Tilt-Up and Pre-Cast Building Construction

4.16 Tilt-Up construction is planned and carried out in accordance with good engineering Practice.

Excavations

Instructions of a Professional Engineer

4.17.1 All excavation work shall be carried out in accordance with the written instructions of a professional engineer where

(1) the excavation is more than 6 m deep,

(2) timber shoring is used in excavations exceeding 3.7 m in width, or

(3) improvements or structures adjacent to the excavation could endanger persons, or

(4) the excavation is subjected to vibration or hydrostatic pressure.

Certification by a Professional Engineer

4.17.2 The written instructions required by section 4.17.1 shall be signed and certified by the professional engineer and be available at the workplace. They shall describe the supporting or sloping requirements and indicate the sub-surface conditions likely to be encountered.

Manager's Requirement

4.17.3 The manager must ensure that no person enters an excavation over 1.2 m in depth, unless

(a) immediately before the person is allowed to enter the excavation,
  (i) a qualified person inspects the excavation, and
  (ii) any hazards identified by the qualified person under subparagraph (i) are made safe, and

(b) at least one of the following has been done in the excavation:
  (i) the sides have been sloped to a safe angle not exceeding 3 horizontal to 4 vertical;
  (ii) the sides have been supported in accordance with the minimum requirements specified in Tables 4-2 and 4-3;
  (iii) the sides have been sloped or supported in accordance with the written instructions of a professional engineer.

Sloping or Combination of Sloping and Shoring

4.17.4 The manager must ensure that when using sloping, or a combination of sloping and shoring, the protection provided to a person is, at the least, equivalent to that of meeting the minimum code requirements for the overall depth of the excavation.

Use of Lumber for Shoring and Timbering

4.17.5 Lumber used for shoring and timbering shall be #2 grade and better, and species to be limited to the following groups: Douglas Fir-Larch, Hem-Fir, Spruce-Pine-Fir or Coast Sitka Spruce. All lumber shall be graded to the National Lumber Grades Authority Rules or other grading rules approved by the chief inspector.

Shoring Contact With Faces of Excavation

4.17.6 Shoring or manufactured or prefabricated support systems shall be installed in firm contact with the faces of the excavation. Any voids shall be backfilled or blocked.

Hydraulic or Pneumatic Jacks

4.17.7 Hydraulic or pneumatic jacks shall have devices which maintain the jacks at their installed length in the event of a loss of internal pressure.

Steel Trench Jacks

4.17.8 Steel trench jacks, with minimum equivalent sizes as shown below, may be substituted for timber struts:

Diameter	Nominal Strut Size Inches (mm)	Nominal Pipe Inches (mm)
(38)	4×4 (89×89)	1.5 Standard
(50)	4×6, 6×6 (89×140, 140×140)	2.0 Standard
(76)	6×8, 8×8 (140×184, 184×184)	3.0 Standard

Use of Plywood

4.17.9 Plywood may be substituted for 50 mm (2") shoring elements provided that

(1) the plywood is not less than 19mm (3/4") in thickness,

(2) the trench is not over 2.7 m in depth,

(3) uprights are installed at no more than 600 mm center to center, and

(4) struts do not bear directly on to plywood

Ladder to Be Kept

4.17.10 A ladder shall be kept in the immediate area of persons working in any excavation over 1.2 m deep.

Requirements Prior to Starting Excavation or Drilling

4.17.11 Prior to starting excavating or drilling, the location of underground utility services in the area shall be accurately determined to ensure persons are not endangered.

Pointed Tools

4.17.12 Pointed tools shall not be used to probe for underground gas and electrical services.

Surrounding Equipment and Objects

4.17.13 Trees, utility poles, rocks and similar objects near an area to be excavated, shall be removed or secured before excavation is commenced.

Excavated Material

4.17.14 Excavated material shall be kept back a minimum distance of 1 m from the edge of any trench excavation less than 3.6 m wide. In any other excavation, the minimum distance shall be 1.5 m.

Danger of Persons Falling Into Excavation

4.17.15 Where there is a danger of persons falling into an excavation, it shall be covered, or standard guardrails or barriers shall be placed along the exposed sides.

EXCAVATIONS OVER 1.2 M IN DEPTH
TABLE 4-2
SIZE AND SPACING OF MEMBERS (IMPERIAL UNITS)

UPRIGHTS			WALES		STRUTS			
Trench Depth (feet)	Minimum Dimension (inches)	Maximum Spacing (feet)	Minimum Dimension (inches)	Maximum Vertical Spacing (feet)	Min. Dimension (inches)		Maximum Spacing	
					Trench Width up to 6'	Trench Width 6 - 12'	Vertical (feet)	Horizontal (feet)
Hard and Solid Soils								
4-10	2×10	6	4×6	4	4×4	6×6	4	6
10-15	2×10	3	6×6	4	4×6	6×8	4	6
Soils Likely to Crack or Crumble								
4-10	2×10	3	4×6	4	4×6	6×6	4	6
10-15	2×10	3	6×8	4	6×6	6×8	4	6
Soft, Sandy, Filled or Loose Soils								
4-10	2×10	Close tight	6×8	4	6×6	6×8	4	6
10-15	2×10	Close tight	8×8	4	6×8	6×8	4	6

NOTES:

(1) Wales may be omitted in trenches not exceeding 8 feet (2.44 m) in depth provided the soil is sufficiently hard and solid to safely permit wale deletion and the trench is not in proximity to previously excavated ground.

(2) At least 2 struts shall be installed in each vertical plane where struts are required.